

OMNYX / THERMYNX — Model Evaluation: Final Combined Report

One document, everything: decision · evidence (local + cloud) · hardware · deployment. Real test on live Unicharm data. Judge = `qwen2.5:32b`. Scores 1–5, **pass = 4+**. Dates: 2026-06-03 (eval) · 2026-06-05 (in-app golden-eval filter). Everything **on-prem** (cloud used for testing big models only). Source files: REAL_DATA_MODEL_EVAL.md, OPENROUTER_MODEL_EVAL.md.

1. DECISION — deploy these (all non-Chinese)

Job	Model	Params	Maker	Country
Planner	mistral-small3.2	24B	Mistral AI	FR FR
Executor	mistral-small3.2	24B	Mistral AI	FR FR
NL→SQL	mistral-small3.2 + guardrails	24B	Mistral AI	FR FR
Validator	phi4	14B	Microsoft	us US
Narration / Text	phi4	14B	Microsoft	us US
RAG	phi4	14B	Microsoft	us US
Vision	llama3.2-vision	11B	Meta	us US
Embeddings	nomic-embed-text	~0.1B	Nomic AI	us US

4 models, ~32 GB total → fit together on the 48 GB box. No upgrade needed.

2. KEY DECISIONS TO TAKE

#	Decision	Why
1	Deploy mistral-small3.2 + phi4 (+vision +nomic)	Best non-Chinese, in-app-working set
2	Keep planned 48 GB box	Whole team fits (~32 GB); no upgrade
3	Don't use 70B/120B	Cloud test: no quality gain, slower
4	Exclude Chinese models (qwen/deepseek/qwq)	Policy
5	Treat NL→SQL as #1 risk	All models weak → fix guardrails/retry
6	Final sign-off: full-depth re-run + FP8 latency test	This was a quick ranking run
7	Rotate OpenRouter API key	Was shared during testing

3. HOW WE TESTED

Item	Detail
Data	Real Unicharm MySQL telemetry + real pgvector document corpus
Jobs (7)	Planner, Validator, Executor, NL→SQL, RAG, Narration, Embeddings
Check 1 (hard rule)	valid JSON / runnable SQL / grounded work order
Check 2 (judge AI)	qwen2.5:32b rates quality 1–5; pass = 4+
Local	≤32B models on the 20 GB box (Ollama Q4)
Cloud	70B/120B via OpenRouter — quality only (speed ≠ our hardware)
Final filter	in-app golden eval (2026-06-05) drops models that score well but break live

4. EVIDENCE — local results (≤32B, on our hardware)

Avg score (pass-rate). **Bold** = **deployed**.

Model	Planner	Validator	Executor	NL→SQL	RAG	Narration
mistral-small3.2	3.0	5.0	3.0	2.8 (best non-CN)	4.4	4.5
phi4	3.0*	5.0	3.0	2.0	4.4	4.5
gpt-oss:20b	3.5	5.0	3.0	3.2	4.4	4.0
gemma3:27b	4.0	5.0	2.0	2.4	4.4	4.5
qwen2.5:14b cn	3.5	5.0	3.5	2.6	4.4	4.5
qwen2.5:32b cn	3.5	5.0	3.0	2.5	4.4	4.5
llama3.1:8b	3.0	5.0	2.5	2.5	4.2	3.0
deepseek-r1:32b cn	3.0 (75s)	2.5	1.5 (434s)	0.0	4.4	err
qwq:32b cn	4.0	5.0	2.5 (226s)	0.0	4.4	4.0
qwen2.5-coder:32b cn	—	—	—	2.25	—	—
Embeddings: nomic = mxbai	—	—	—	—	—	both 5.0

* phi4 had 1 malformed plan on the reduced run; strong elsewhere → deployed for Validator/RAG/Narration.

5. EVIDENCE — cloud results (big models, quality-only)

Model	Params	Planner	Validator	Executor	RAG	Narration
gpt-oss-120b	120B	4.0 (errored 2/3)	5.0	2.33	4.4	4.0
llama-3.3-70b	70B	3.67	5.0	2.33	4.4	4.5
qwen-2.5-72b cn	72B	3.0	5.0	3.33	4.25	4.0
deepseek-r1-distill-70b cn	70B	0 (fail)	1.0	1.0	4.4	4.5
gpt-oss-20b	20B	4.0	5.0	2.33	4.5	4.0

Verdict: bigger did NOT win — 70B/120B tie at best, run slower, 120B timed out on planning. (NL→SQL not run on cloud — backend SQL path can't route cloud models.)

6. WHY THE RAW WINNERS AREN'T DEPLOYED

Model	Note
gpt-oss:20b	Good & fits , won 3 jobs — but reasoning model → empty answers under our token caps; failed golden eval . Fixable (raise token budget + thinking-aware streaming).
gemma3:27b	Won planner but broke tool-calling (2.0) ; failed golden eval; 27B blocks co-residency.
qwen / deepseek / qwq	Chinese-origin — excluded by policy . (qwen2.5:32b used only as offline judge.)

7. HARDWARE — servers & fit

	Current (test)	Future (production)
GPU VRAM	20 GB (RTX 4000 Ada)	48 GB
RAM	32 GB	48 GB
Serving	Ollama (Q4)	vLLM (FP8)
Loadable at once	1 chat model	all 4 models

What fits in the 20 GB box today (one chat model + tiny embedder):

Loadout	VRAM	OK?
phi4 + nomic	~9 GB	✓
mistral-small3.2 + nomic	~15 GB	✓
llama3.2-vision + nomic	~8 GB	✓
phi4 + mistral together	~24 GB	✗ swap
70B / 120B	42 / 65 GB	✗

Model size → fit: 11–24B ✓ ideal · 27–32B ✓ alone only · 70B needs 80 GB (FP8) · 120B can't run on-prem.

Hardware tiers: A = **48 GB (recommended, proven best)** · B = 80 GB (optional, no gain) · C = 160 GB (not justified).

20 GB box vs 48 GB box — pros / cons / limitations

Aspect	20 GB box (current / test)	48 GB box (future / production)
Models loaded at once	1 chat model + tiny embedder	All 4 models together
Multi-agent (Planner+Validator+Executor)	✗ run one at a time	✓ run in parallel
Model swapping	⚠ yes — reload delay when role changes	✓ none — all resident
Precision / quality	Q4 (lower)	FP8 (higher quality)
Biggest model	≤32B (alone, tight)	≤32B easily; 70B only 4-bit alone
Production-ready?	✗ dev/screening only	✓ yes
Cost	already owned	new GPU spend

Pros — 20 GB box: already have it; free; runs **all our chosen models** (one at a time); perfect for testing, screening, and the eval we just did.

Cons / limitations — 20 GB box: only **one chat model fits at a time**, so a live agent flow (plan → validate → execute) must **swap models between steps** (seconds of delay each switch); **no concurrency**; tight for 27–32B; reasoning models very slow; **cannot host 70B/120B**. → **Not suitable for production multi-agent load**.

Pros — 48 GB box: the **whole 4-model team stays loaded** (~32 GB, ~16 GB spare) → **no swapping, agents run together**, lower latency; **FP8 = better quality** than the Q4 we tested; production-grade; headroom for concurrent users.

Cons / limitations — 48 GB box: new GPU cost; still **can't run a 70B at FP8** (only 4-bit, and then *alone*) or a 120B — those need an **80 GB** card. (Per the eval, that's fine — bigger models gave no benefit.)

Verdict: keep the **20 GB box for dev/testing**; deploy production on the **48 GB box** — it removes the swapping limit and runs the full agent team at higher precision. No need to go beyond 48 GB.

8. DEPLOYMENT — live config (`backend/app/config.py`)

Job	Setting	Model
Default	<code>OLLAMA_DEFAULT_MODEL</code>	phi4
Text / Narration	<code>OLLAMA_MODEL_TEXT</code>	phi4
Tool / Executor	<code>OLLAMA_MODEL_TOOL</code>	mistral-small3.2
NL→SQL	<code>OLLAMA_MODEL_SQL</code>	mistral-small3.2
Planner	<code>OLLAMA_MODEL_PLANNER</code>	mistral-small3.2
Validator	<code>OLLAMA_AUDITOR_MODEL</code>	phi4
RAG	<code>OLLAMA_MODEL_RAG</code>	(empty)→phi4
Vision	<code>OLLAMA_VISION_MODEL</code>	llama3.2-vision
Embeddings	pipeline	nomic-embed-text

Tuning: AGENT_MAX_STEPS=8 · NL_QUERY_LLM_TIMEOUT=40s · CHAT_TIMEOUT=60s · VISION_TIMEOUT=90s. **Guardrails:** NL→SQL validator + deny-list + LIMIT cap · work-order numeric-citation guard (never auto-persists).

9. BOTTOM LINE

Question	Answer
Best models	mistral-small3.2 (24B) + phi4 (14B) + llama3.2-vision + nomic-embed-text — all non-Chinese
Best hardware	Planned 48 GB box (FP8) — fits the team (~32 GB); no upgrade
Need 70B/120B?	No — bigger ties/loses; would need 80 GB+
Biggest risk	NL→SQL — fix with guardrails/retry, not a bigger model
Revisit later	gpt-oss:20b — with higher token budget + thinking-aware streaming
Cross-check	Confirms earlier synthetic pick on real data — safe to lock in

One line: run **mistral-small3.2 + phi4** (+ vision + nomic) on the **planned 48 GB FP8 box** — best-grade, best-fit, all-non-Chinese, no extra hardware needed.

Detailed companions: MODEL_FIT_VERDICT.md (decision) · ONPREM_HARDWARE_SIZING.md (hardware/deploy) · REAL_DATA_MODEL_EVAL.md & OPENROUTER_MODEL_EVAL.md (raw per-case

tables).